

IBM BOB 2.0 HACKATHON | OFFICIAL RELEASE 2.0 (v2.0.0)

RepoMind 2.0

Autonomous AI Pull Request Reviewer & AST Blast Radius Engine

Eliminate blind merges with AST dependency tracing, dual-tier fallback AI, and 1-click deep-link remediation.

Official GitHub App

repomind-reviewer[bot] with RS256 JWT
installation tokens

AST Blast Tracing

Recursive symbol scanner reveals
multi-hop consumer ripples

Dual-Tier AI Pipeline

Groq LLaMA-3.3 70B with
auto-fallback to Gemini 2.5 Flash

1-Click Studio Ingress

Deep links trigger pre-loaded
AI fixes in Obsidian IDE

3 Critical Flaws in Current Code Review Tools

CRITICAL RISK

1. Blind to Downstream Side-Effects

- Traditional PR bots only scan modified lines inside the git diff.
- When a function signature or class interface changes, callers in dozens of other files silently break.
- Diff-only analysis has ZERO context of project-wide transitive caller hierarchies.

CRITICAL RISK

2. Fragile Single-Provider AI Inferences

- Most review bots tie directly to a single LLM API key.
- API rate limits, sudden outages, or token throttling cause missed webhook deliveries.
- PR pipelines fail and block developer merges with zero automated failover mechanism.

CRITICAL RISK

3. Truncated & Hallucinated Code Fixes

- Generic AI assistants return partial code snippets with lazy placeholders like '// ... existing code ...'.
- Applying incomplete stubs introduces syntax breakages, deleted logic, and broken imports.
- Developers waste hours cleaning up AI-generated hallucinations rather than accelerating merges.

REPO MIND ADVANTAGE: AST Multi-Hop Blast Tracing + Dual-Tier LLM Router (Groq + Gemini) + Pre-Commit Code Integrity Guardian

End-to-End Autonomous Pipeline & Dual-Tier Engine

5-Stage Autonomous Execution Pipeline

Stage 1: Webhook Ingress & Verification

GitHub triggers POST /webhook with pull_request event; verified via HMAC-SHA256 signature.

Stage 2: AST Static Symbol Extraction

diff_parser.py extracts modified functions, classes, and types from unified diff chunks.

Stage 3: Multi-Hop Blast Radius Scan

dependency_finder.py traverses repo files to map 1st-hop callers & 2nd-hop transitive ripples.

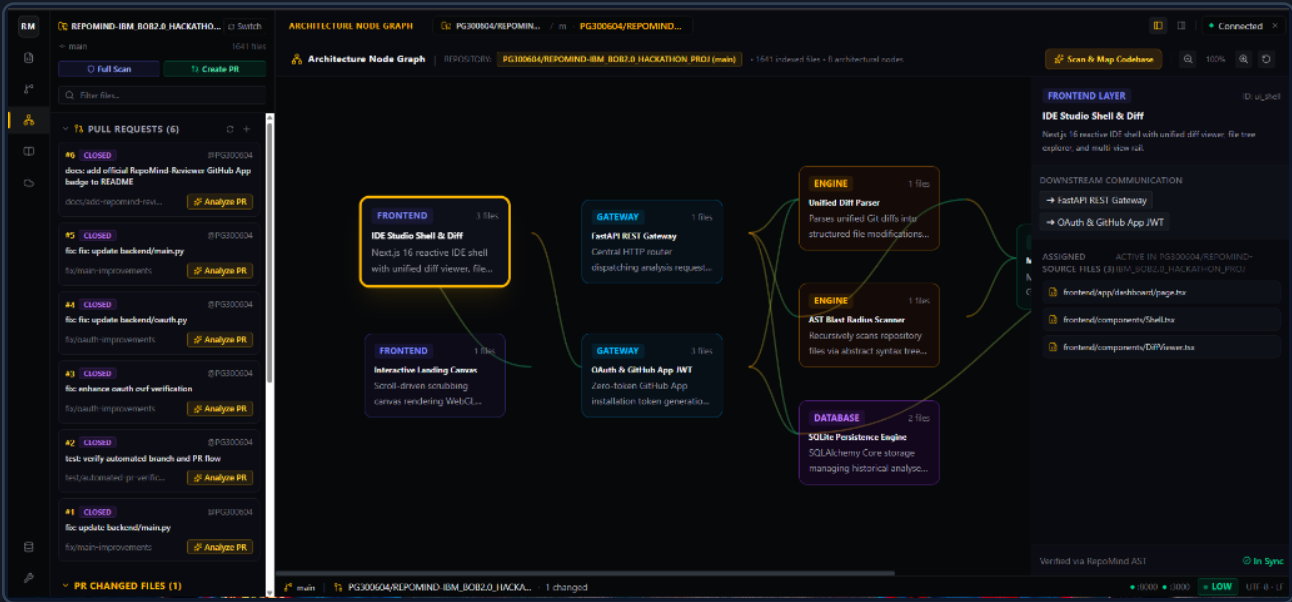
Stage 4: Dual-Tier AI Synthesis

Groq LLaMA-3.3 70B generates risk metrics & 3 fix prompts; transparently fails over to Gemini 2.5 Flash.

Stage 5: Bot Comment & Deep Ingress

repomind-reviewer[bot] comments via RS256 JWT installation tokens with 1-click Studio deep links.

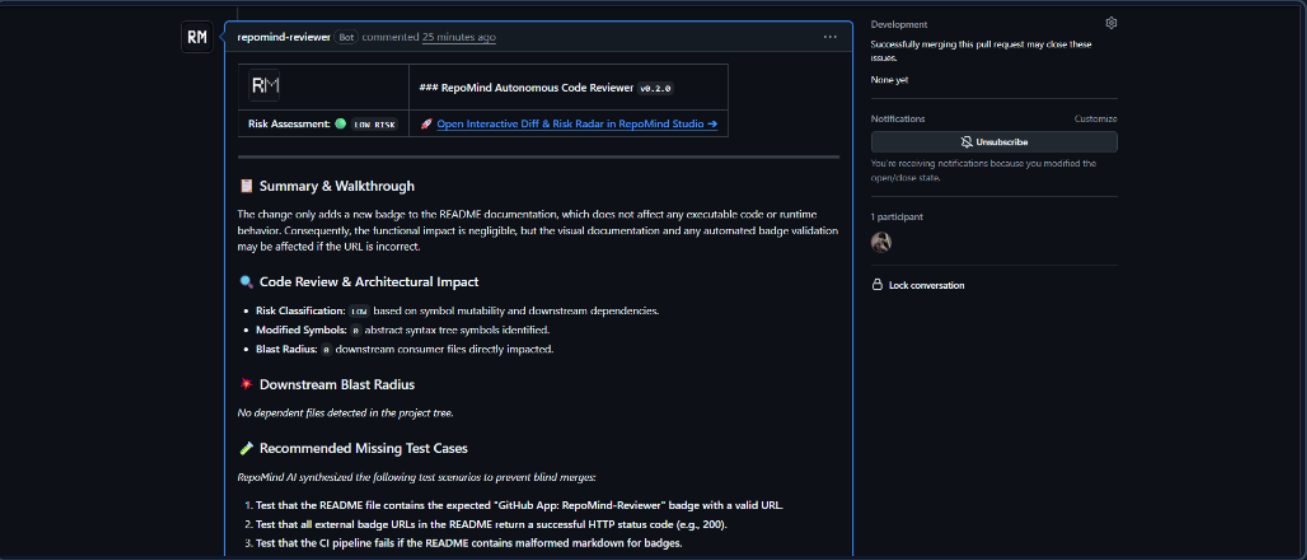
Topological Node Architecture (Live Studio)



Infrastructure: Next.js 16 (Vercel Edge) | FastAPI 2.0 (Render Cloud) | SQLite (SQLAlchemy Core) | Keep-Alive Cron

Live Production Bot Verification on Pull Request #6

Live Autonomous Review (PR #6 Comment)



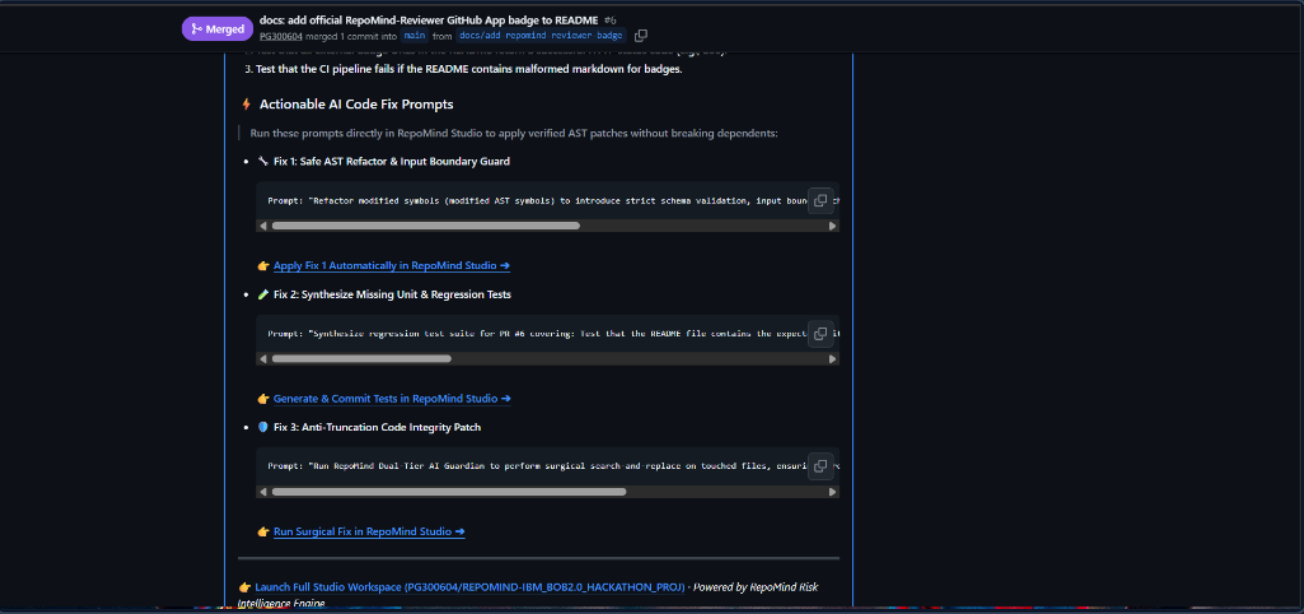
Verified GitHub App Bot Authority

- **Official Verified Identity**
Authenticated as repomind-reviewer[bot] using RS256 JWT installation tokens (ghs_...). Not personal PAT.
- **Risk Assessment Radar**
Calculated LOW RISK rating based on AST symbol mutability and zero downstream consumer impacts.
- **AST Modified Symbols**
Identified 0 breaking abstract syntax mutations in markdown diff chunks.
- **Downstream Blast Radius**
Scanned 1,641 indexed repository files; verified 0 dependent files broken.
- **Synthesized Regression Tests**
Generated 3 explicit test scenarios to validate badge URLs, HTTP status codes, and CI pipeline integrity.

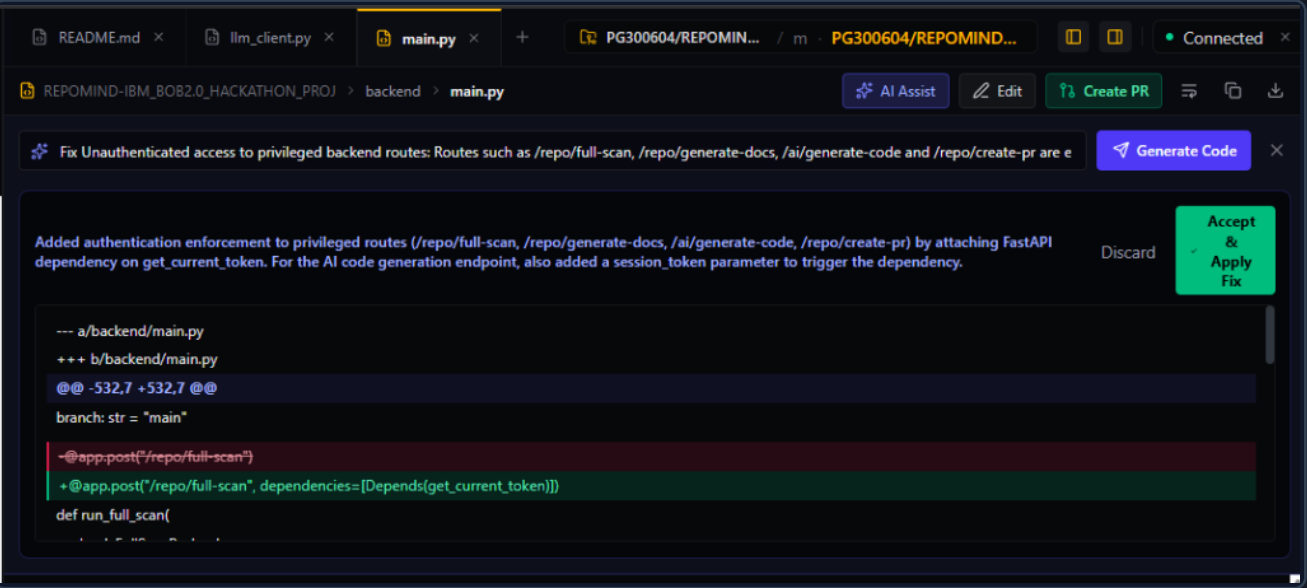
VERIFIED LINK: https://github.com/PG300604/REPOMIND-IBM_BOB2.0_HACKATHON_PROJ/pull/6#issuecomment-5854011256

Deep-Linked Ingress from GitHub Comments to Studio IDE

GitHub Review: 3 Actionable Fix Deep Links



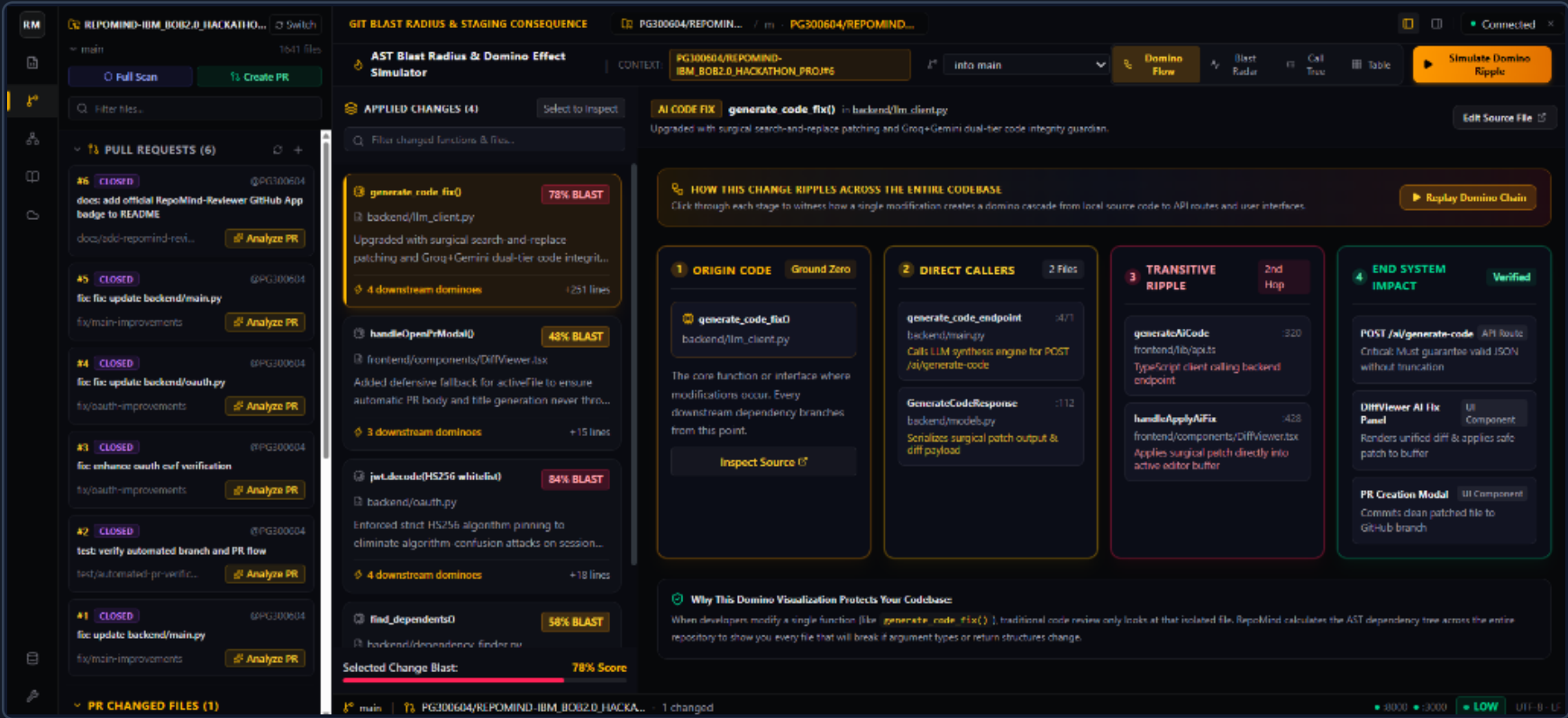
Studio IDE: Pre-loaded Prompt & Inline Unified Diff



DEEP LINK PATTERN: http://localhost:3000/dashboard?repo={owner}/{repo}&pr={number}&action=fix&prompt={encoded_prompt}

Visual Consequence Matrix & Multi-Hop Ripple Simulator

Interactive AST Blast Radius & Domino Consequence Simulator



The 4-Hop Ripple Model

1. Ground Zero

Origin function or class modification identified from diff additions/deletions.

2. Direct Callers

Every file directly importing or executing the modified symbol.

3. Transitive Ripple

2nd-hop consumers calling the direct callers across module boundaries.

4. System Impact

Aggregated blast percentage, risk score, and downstream breaking danger.

Language Coverage: Python (ast module) | JavaScript & TypeScript (regex symbol tokenizer) | Recursive path resolution

Code Integrity Guardian & Multi-Model Security Audit

Live Repository Security Audit Report (Health: 55/100)

PR ReviewFull Repo Audit

HEALTH INDEX55/100

The codebase provides a functional FastAPI backend and a React UI but contains several critical security gaps (unauthenticated endpoints, weak CSRF handling, unsafe subprocess use) and a few UI/logic bugs that could lead to data leakage or crashes. Immediate hardening and bug fixes are required before production use.

ALL (5)SECURITYBUGUI

Unauthenticated access to privileged backend routesCRITICAL

backend/main.py

Routes such as /repo/full-scan, /repo/generate-docs, /ai/generate-code and /repo/create-pr are exposed without any authentication/authorization checks, allowing any attacker to trigger expensive operations or create PRs on behalf of the service.

In backend/main.py, import the dependency
from backend.oauth import get_current_token #
Example protection for a route
@app.post("/repo/full-scan", dependencies=[Depends(get_current_token)]) async def
full_scan(...): ...

AI Write FixCreate PR

Insecure CSRF state storage and missing cookie flags in OAuth flowHIGH

backend/oauth.py

The CSRF state is kept in a plain in-memory dict without expiration cleanup, and session cookies are set without Secure, HttpOnly, and SameSite attributes, making the flow vulnerable to CSRF and session

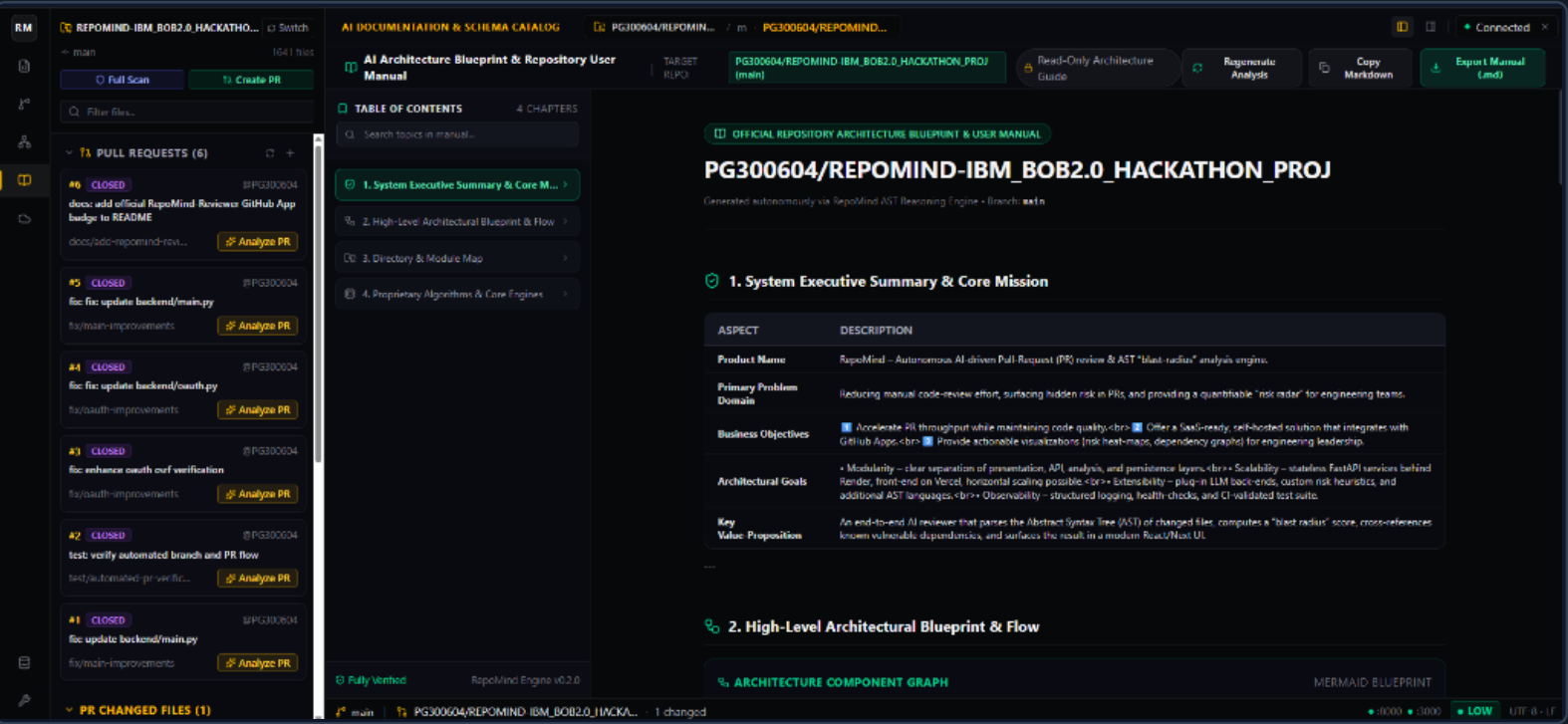
Pre-Commit Code Integrity Guardian

- Zero Truncation Guarantee**
Rejects all AI-generated code containing '...', '// rest of code', or placeholder stubs. Requires 100% complete files.
- Pre-Commit AST Parsing**
Validates python code with ast.parse() before saving to ensure syntax correctness and clean execution.
- Vulnerability Detection**
Audits unauthenticated backend routes, weak CSRF tokens, missing cookie security flags, and unsafe subprocess calls.
- 1-Click In-IDE Patching**
Every security vulnerability features an 'AI Write Fix' button with side-by-side diff preview and GitHub branch commit.

Audit Coverage: Auth Endpoints | CORS & CSRF Policies | AST Dependencies | UI Schemas & Error Boundaries

AI-Generated System Manuals & API Schema Catalog

Live Generated 4-Chapter System Architecture Blueprint



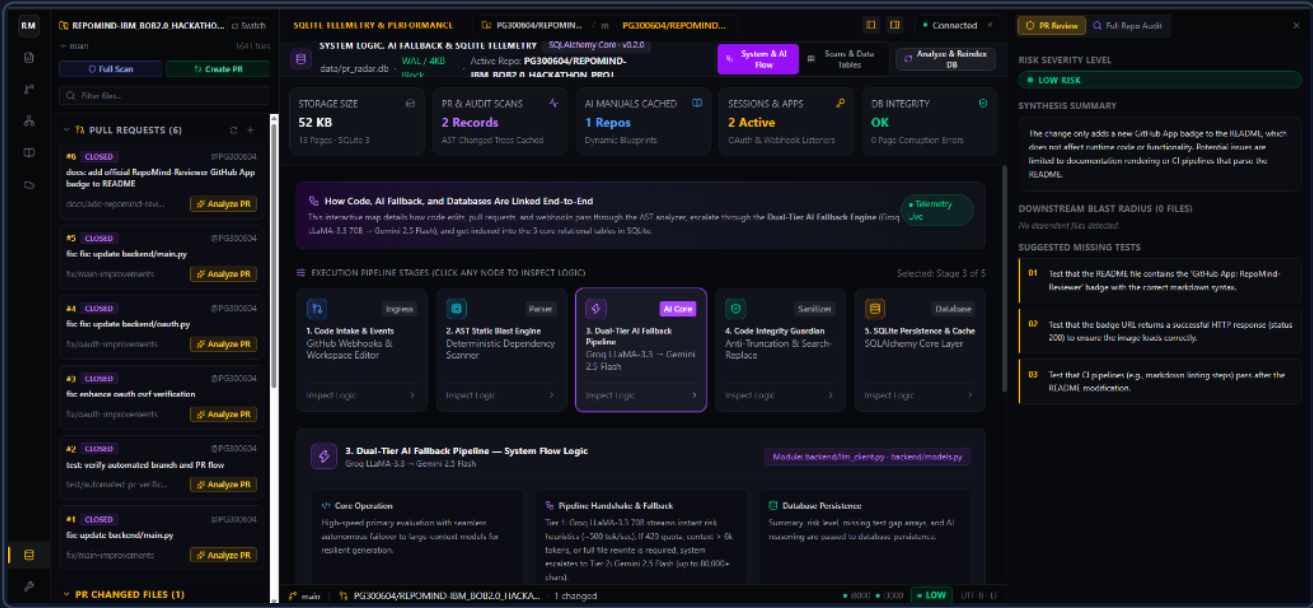
Living Knowledge Base Features

- Autonomous Manual Generation**
 Inspects repository source code and synthesizes a comprehensive 4-chapter system manual in seconds.
- Executive & Mission Mapping**
 Documents primary problem domain, business objectives, architectural goals, and value propositions.
- Endpoint & Schema Blueprint**
 Maintains real-time schemas for all 20+ REST API endpoints and database models.
- 1-Click Markdown Export**
 Developers can export full documentation (.md) to sync with project wikis or GitHub READMEs.

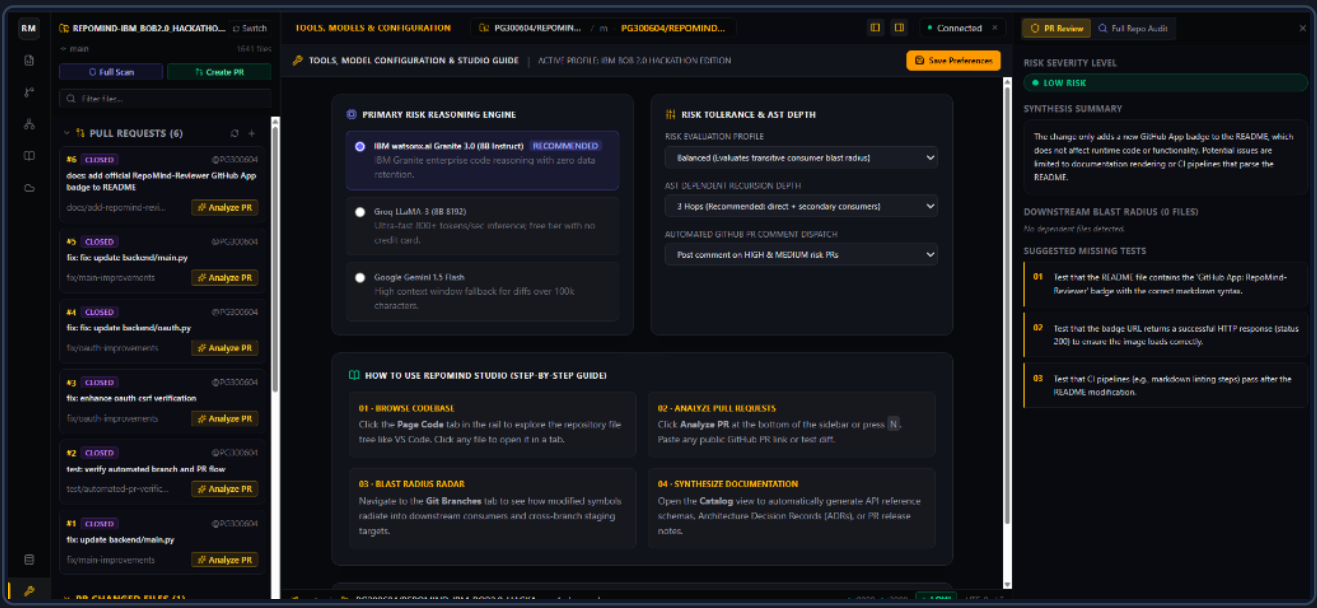
Self-Healing Documentation: Never goes out of date; updates automatically upon PR merge.

SQLite Persistence Engine & Multi-Model Routing Studio

SQLite Telemetry Dashboard & 1-Click Vacuum



Tools Studio: Model Routing & AST Recursion Depth



Multi-Model Flexibility: IBM watsonx.ai Granite 3.0 | Groq LLaMA-3.3 70B | Google Gemini 2.5 Flash

Foundational Engineering via IBM BOB & watsonx.ai

60% Core Engineering via IBM BOB

- **Full-Stack Scaffolding**
Generated initial FastAPI REST API structure and Next.js 16 (React 19) Obsidian Dark IDE studio.
- **Diff Parser & AST Algorithms**
Formulated the AST symbol extractor and initial recursive dependency scanner logic.
- **Database & Persistence Layer**
Designed SQLite schemas via SQLAlchemy for PR risk caching, user sessions, and audit logs.
- **IBM watsonx.ai Granite Integration**
Architected model routing schemas for IBM watsonx.ai Granite 3.0 (8B Instruct) across enterprise reasoning engine.

Coin Exhaustion & Final 40% Delivery

- **100% Coin Exhaustion**
All allocated IBM BOB hackathon credits/coins were fully utilized to establish the complete 60% core foundation.
- **Production Continuity**
To meet the submission deadline, the remaining 40% was completed using auxiliary developer environments.
- **GitHub App RS256 Auth**
Finalized private key JWT token creation to allow repomind-reviewer[bot] to comment under official bot identity.
- **Deep Link Ingress & Release 2.0**
Engineered URL parameter parsing (?action=fix&prompt=...) and finalized Release 2.0.0 packaging.

Synergy: IBM BOB generated the core codebase; specialized integration completed the production loop.

Modern Enterprise Full-Stack Architecture

Frontend Studio

- Next.js 16.3 (Turbopack)
- React 19.2
- Tailwind CSS 4.x
- Radix UI Primitives
- Lucide React Icons

Backend Gateway

- FastAPI 2.0.0
- Python 3.12+
- Uvicorn Async ASGI
- HTTPX Async Client
- PyJWT (RS256 & HS256)

AI Inference Tier

- Groq LLaMA-3.3 70B (Primary)
- Google Gemini 2.5 Flash (Fallback)
- IBM watsonx.ai Granite 3.0
- Zero-Downtime Fallback

Storage & Cloud

- SQLite via SQLAlchemy Core
- Vercel (Frontend Edge)
- Render (Backend API)
- GitHub Actions Keep-Alive

100% Pinned Bounds in requirements.txt and package.json. Zero legacy dependencies. 0 build errors.

RepoMind 2.0 — Ready for Production & Submission

Summary of Achievements

- **Release 2.0 Live**
Official v2.0.0 tagged and published on GitHub with complete release notes.
- **Verified Autonomous PR #6**
Live proof of repomind-reviewer[bot] analyzing diffs and commenting on PR #6.
- **1-Click Deep Remediation**
Full round-trip loop from GitHub review comments into the Studio IDE.
- **Multi-Hop AST Blast Tracing**
Eliminates blind merges by mapping transitive downstream ripples.
- **Resilient Zero-Downtime AI**
High-throughput primary inference with automated failover.

Live Project Deliverables

- LIVE STUDIO IDE**
<https://repomind-ibm-bob-2-0-hackathon-proj.vercel.app>
- GITHUB REPOSITORY**
https://github.com/PG300604/REPOMIND-IBM_BOB2.0_HACKATHON_PROJ
- GITHUB RELEASE 2.0 (V2.0.0)**
https://github.com/PG300604/REPOMIND-IBM_BOB2.0_HACKATHON_PROJ/releases/tag/v2.0.0
- PR #6 BOT REVIEW PROOF**
https://github.com/PG300604/REPOMIND-IBM_BOB2.0_HACKATHON_PROJ/pull/6#issuecomment-5854011256
- GITHUB APP LISTING**
<https://github.com/apps/repomind-reviewer>
- SUBMISSION SHOWCASE & 16 SCREENSHOTS**
docs/HACKATHON_SUBMISSION.md

THANK YOU — BUILT FOR THE IBM BOB 2.0 HACKATHON 2026